

Supplementary Table S8. Meth3D-Net V6 Cross-Dataset Validation Summary — Three-Layer Independent Validation in Two Cohorts

Layer A = Variance enrichment. Layer B = Epigenomic chromatin instability (CT) replication. Layer C = Direction concordance. All datasets are independent of V6 training data. p-values below scipy float precision floor reported as $p < 2.2 \times 10^{-300}$.

Layer	Metric	Cohort 1: GSE85212 MB (n = 763; Illumina 450k)	Cohort 2: TCGA PanCancer (n = 9,854; 33 cancer types)	Interpretation	Validation conclusion
A	Fold enrichment (high- $ \Delta\beta \geq 0.30$ vs low- $ \Delta\beta < 0.30$) Mann-Whitney U test	1.62x median: 0.02155 vs 0.01329 n = 85,134 probes	1.36x median: 0.0375 vs 0.0276 n = 105,451 probes	Both cohorts show significant enrichment of V6-predicted high- $ \Delta\beta $ positions among high-variance CpGs in independent experimental data. $p < 2.2 \times 10^{-300}$ in both cohorts (below scipy float precision floor).	Validated (both cohorts)
A	Spearman r ($ \Delta\beta $ magnitude vs methylation variance) continuous correlation	r = +0.039 p = 2.5×10^{-30} n = 85,134 probes	r = +0.271 p < 2.2×10^{-300} n = 105,451 probes	Monotonic positive relationship between V6 predicted methylation variability and observed variance in independent cohorts. Stronger r in TCGA reflects greater pan-cancer methylation variance across 33 cancer types.	Validated (both cohorts)
B	Spearman r (CT instability z-score vs methylation variance)	r = +0.039 p < 2.2×10^{-300} 2,145,652 CT windows 50 kb resolution	r = +0.013 p = 4.1×10^{-81} 2,145,652 CT windows 50 kb resolution	CT instability scores computed from V6 MB prediction residuals correlate significantly with independent cohort methylation variance. Smaller r in TCGA reflects resolution mismatch (50 kb MB-derived windows vs probe-level TCGA variance across diverse tissue types).	Validated (both cohorts)
C	Direction concordance (%) (V6 predicted $\Delta\beta$ direction vs observed differential methylation direction)	58.4% Group3 vs WNT n = 84,394 probes (50% = chance)	40.8% Tumour (n=8,695) vs normal (n=671) n = 105,451 probes (50% = chance)	MB: > 50% concordance confirms V6 directional accuracy above chance. TCGA: < 50% concordance is mechanistically EXPECTED — positions hypermethylated in IMR90 vs H1 ESC are preferentially hypomethylated in tumour vs normal (global cancer hypomethylation; Feinberg & Vogelstein 1983).	Validated (MB: > chance; TCGA: expected inverse)
C	Spearman r (V6 $\Delta\beta$ direction vs cohort differential methylation direction)	r = +0.025 p = 2.1×10^{-13} n = 84,394 probes	r = -0.248 p < 2.2×10^{-300} n = 105,451 probes	MB: positive r confirms V6 directional predictions replicate in independent primary tumour data. TCGA: negative r quantifies global cancer hypomethylation (magnitude -0.248); both values reject null of r = 0.	Validated (MB: positive; TCGA: expected negative)
C	Near-zero $ \Delta\beta $ probes ($ \Delta\beta < 0.01$; effectively invariant CpGs)	26.2% (Group3 vs WNT) Down from 75.0% (SHH vs WNT) – 2.9-fold reduction	12.0% (12,569 / 105,451 probes) Greater pan-cancer variance in TCGA vs single MB cohort	Reduction from 75.0% (SHH-WNT) to 26.2% (Group3-WNT) confirms that Group3 activates a substantially larger fraction of V6-predicted variable positions, consistent with the greater molecular divergence between these subgroups. TCGA 12% reflects genuine pan-cancer methylation variability.	Consistent with V6 predictions

Layer colour key:

Layer A	Variance enrichment — V6-predicted high- $ \Delta\beta $ CpGs enriched among high-variance positions in independent cohorts
Layer B	CT instability replication — epigenomic chromatin instability scores correlate with independent methylation variance
Layer C	Direction concordance — V6 $\Delta\beta$ direction matches observed differential methylation direction in independent cohorts

Supplementary Table S8 (continued) — Part 2: Corrected p-value Reference (resolving corrupted Unicode in original Table 3)

The original Table 3 contained corrupted Unicode exponent characters in several cells. This section provides the correct values with the source of each corruption identified.

Metric / location in original Table 3	Original corrupted string	Problem	Corrected value
Layer A, MB cohort — fold enrichment p-value	p<1x1-12	Missing base 10; not a valid number	p < 2.2x10-300 (below scipy float floor)
Layer A, TCGA — Spearman r and p-value	p<10.271 (p=0.000; n=105,451 probes)#xD7;100.271 ...	Garbled Unicode: r value mixed into p	r = 0.271; p < 2.2x10-300
Layer A, TCGA — p=0.000 (multiple occurrences)	p=0.000	Float rounding artefact; meaningless	p < 2.2x10-300
Layer B, TCGA — Spearman r(CT-z, variance)	0.013 (p=4.1x10-5) — correct in some versions	No corruption (verify source)	r = 0.013; p = 4.1x10-81 (n=2,145,652 windows)
Layer C, MB — Spearman r direction	0.025 (p=2.1x10-13)	No corruption (correct value)	r = +0.025; p = 2.1x10-13
Layer C, TCGA — Spearman r direction	-0.248 (p=0.000)#xD7;10-0.248 ...	Unicode corruption + p=0.000 artefact	r = -0.248; p < 2.2x10-300

Notes

1. All p-values below scipy float precision (approx. 2.2x10-300) are reported as p < 2.2x10-300; actual values are smaller.
2. TCGA Layer C negative Spearman r (-0.248) is mechanistically expected: positions hypermethylated in differentiated IMR90 vs pluripotent H1 ESC are preferentially hypomethylated in cancer vs normal tissue — the global cancer hypomethylation phenomenon first described by Feinberg & Vogelstein (1983, Nature 301:89-92).
3. GSE85212: Cavalli et al., Cancer Cell, 2017. TCGA PanCancer: Liu et al., Cell 173:400-416, 2018. V6 training data: GSE19418 (Ziller et al., Nature 500:477-481, 2013).
4. CT = epigenomic chromatin instability score; |Δβ| = absolute V6-predicted methylation difference (IMR90 minus H1 ESC beta value).